

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	
Project Name	OptiCrop
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Crop Recommendation	Predicts the best crop from 7 soil/climate inputs	ML / app.py	Done	2
2	Prediction API	Real-time Flask endpoint returning the crop	Backend	Done	1
3	Input Form	Interactive form for N,P,K,temp,humidity,pH,rain	Frontend	Done	1
4	Glassmorphism UI	Animated, responsive result presentation	Frontend	Done	0.5
5	Error Handling	Graceful handling of invalid inputs	Backend	Done	0.5
6	Crop Mini-Game	Virtual crop-clicker engagement feature	Frontend	Done	0.5
7	EDA Plots	Univariate/bivariate/correlation visualizations	ML	Done	0.5
8					

Feature Summary:

Metric	Count / Value
Total Features Planned	7
Total Features Implemented	7
Core / Must-Have Features	3

Additional / Nice-to-Have Features	4
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Input form, UI animations, mini-game
2	Backend / Logic	3	Prediction, validation, routing
3	Database / Storage	1	model.pkl, Crop_recommendation.csv
4	API / Integration	1	Flask REST endpoint
5	Security / Authentication	0	Public (no auth)